Simulation of Crop Response to Water and Nitrogen: An Example Using Spring Wheat

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ABSTRACT

A model for simulating the effect of water and nitogen on growth and grain yield of spring wheat is presented. This model attempts to offer a mechanistic integration of water, heat and solute transport, nitrogen transformation, and nitrogen and water uptake for the simulation of crop growth. Calibration and testing of the model is discussed and improvement for the current approach are suggested.

INTRODUCTION

The need to extrapolate experimental crop response to water and nitrogen treatments to locations and growing seasons other than those of the experiments has led to the development of predictive methods. The ultimate target is to make available tools for evaluating management options, and provide guidance for identifying and pretesting research options.

The objective of this work was to develop a conceptual frame to simulate crop growth using a mechanistic description of water, heat and solute transport in the soil, nitrogent transformations, and nitrogen and water uptake. This effort represents an extension of previous work (Stockle and Campbell, 1985). The concepts included in the model have been applied to simulate spring wheat response to water and nitrogen management, but they apply to other grain cereals. Crop growth models accounting for the response to water and nitrogen are available (Watts and Hanks, 1978; Meyer et al., 1979; Baker et al., 1981; Tillotson and Wagenet, 1982; Jones and Kiniry, 1986; Van Keulen, 1987). However, none of these models present a complete and simulataneous mechanistic description of water, solute, and heat transport linked to a reasonably detailed crop growth description.

THE CONCEPTUAL MODEL

A conceptual description of the model, emphasizing mainly the nitrogen related components, and some of the governing equations will be presented in this section. Model components like light interception, photosynthesis, water flux in the soil-plant-atmosphere continuum (based on water potential gradients and resistances), water uptake, and the relationships between these elements have been discussed in an earlier paper (Stockle and Campbell, 1985) and they will not be included here. A more detailed presentation of the model is available elsewhere (Stockle, 1985) and several of the model's components are also discussed by Campbell (1985).

Transport Processes

A numerical approach is used to describe the tranport of mass and energy within the soil-plant-atmosphere system. The proposed method divides the system into a number of layers whose properties are concentrated at nodes, and monitors the transfer of mass and energy through each layer. Storage is assumed to occur only at the nodes. The system is represented by an electric analog (Fig. 1), in a similar fashion to that described by Bristow (1983). Node 0 corresponds to the reference height at which weather parameters are measured, node 1 is located at the zero plane displacement height within the crop canopy, node 2 corresponds to the soil surface, and the remaining nodes, to different depths in the soil profile.

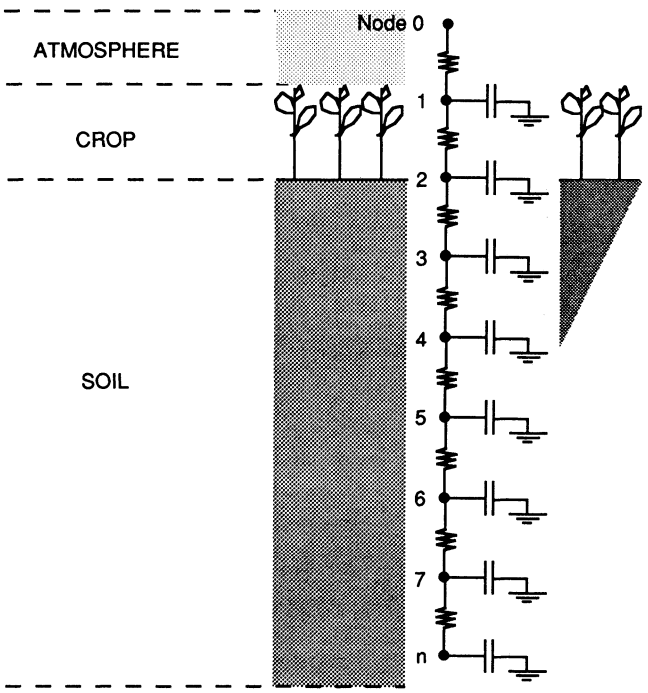


Fig. 1—Electrical analog of heat and water flow in the soil-crop-atmosphere system showing node numbers, resistances and storage locations.

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Water Transport

Infiltration, redistribution, evaporation, and plant uptake (transpiration) are simulated. Deep drainage occurs at the bottom of the profile as a result of a gradient in gravitational potential. The matric potential induced flux is assumed negligible at the bottom of the profile. The top boundary condition is a specified flux, and the bottom, a specified matric potential.

A numerical solution to the network flow problem is used to find water potential and water content changes with time at each node in the soil. The complicating factor is that, for unsaturated flow, the hydraulic conductivity and water capacity are functions of the dependent variable (water content or potential). A method is therefore needed that solves non-linear equations. The Newton Raphson iterative method is used. Details of the solution are described by Campbell (1985).

Heat Transport

A finite difference scheme was used to simulate changes in soil temperature with time. A detailed description of the procedure is given by Campbell (1985). Soil thermal conductivity was assumed to vary with water content and soil texture as given by Campbell (1985). Conductance for the region between the top node (node 0, atmosphere) and the next one down (node 1 at the zero plane displacement height within the crop canopy) is obtained from the boundary layer resistance for turbulent transfer at a surface (Campbell, 1977):

$$K_0 = \rho c_p \, k \, u^*/\ln [(z_r - d - z_H)/z_H] \quad \ldots\ldots\ldots [1]$$

where the friction velocity, u*, is given by

$$u^* = k \, u_r/\ln [(z_r - d + z_M)/z_M] \quad \ldots\ldots\ldots\ldots [2]$$

and ρ is the density of the air (g m^{-3}), c_p is its specific heat (J g^{-1} K^{-1}), u_r is mean windspeed (m s^{-1}) measured at reference height z_r, k is von Karman's constant (generally taken as 0.4), d is the zero plane displacement (m), equal to 0.77 times crop height, z_M is the momentum roughness parameter (m), estimated as 0.13 times crop height, and z_H is the heat roughness parameter (m), estimated as 0.2 times z_M.

A single value of heat conductance (Thom, 1975) is assumed for the zone between node 1 and 2 (Fig. 1) and is given by

$$K_1 = k \, u^* \, (h + z_M - d)/d \quad \ldots\ldots\ldots\ldots\ldots [3]$$

where h is the crop height (m).

The boundary conditions for heat flow are the air temperature at z_r and the soil temperature at the bottom of the soil profile. Heat sources account for radiant and latent heat exchange at nodes 1 and 2 (crop and soil exchange surfaces).

Solute Transport

Nitrogen flow in the bulk soil is obtained by multiplying water flux by the nitrate or ammonium concentration in solution. Convection is the only mechanism included for simulating nitrogen flow in the soil profile, though numerical dispersion substitutes, to some extent, for physical dispersion.

In the case of ammonium, there is need to account for the interaction between liquid and solid phase. For this purpose, it is assumed that the absorbed and solution phase solute concentrations are related by a Langmuir relation (Campbell 1985):

$$X_N = CQN_a/(1 + KN_a) \quad \ldots\ldots\ldots\ldots\ldots\ldots [4]$$

where X_N is ammonium on exchange sites of the solid phase (kg kg^{-1}), N_a is soil solution ammonium concentration (kg kg^{-1}), and C and Q are constant for a given soil and solute (kg kg^{-1}). The total amount of ammonium in the soil (S_{as}) is

$$S_{as} = X_N + wN_a\rho_b \quad \ldots\ldots\ldots\ldots\ldots\ldots [5]$$

where S_{as} is in kg m^{-3}, p_b is soil bulk density (kg m^{-3}) and w is the mass basis water content (kg kg^{-1}). Equations [4] and [5] can be combined and solved for ammonium concentration in solution using the quadratic formula. The resulting equation allows the ammonium concentration in the soil solution in equilibrium with the solid phase to be computed for a given total amount of ammonium in the soil.

Nitrogen Transformations

Microbial nitrogen transformations are assumed to occur in the top 30 cm of the soil. These are net mineralization, nitrification and denitrification, and they are assumed to obey irreversible first order kinetics, as suggested by Mehran and Tanji (1974).

$$dN/dt = -KN \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots [6]$$

where N denotes nitrogen species concentration, dt is time interval and K is a rate constant.

The description of nitrogen transformations follows the scheme proposed by Watts and Hanks (1978). The description of the net mineralization (mineralization -immobilization) of organic matter to ammonium is based on the work of Stanford and Smith (1972):

$$M = [M_0(1 - \exp(-K_M \, \Delta t))] \, F(f_{ps}) \quad \ldots\ldots\ldots\ldots [7]$$

where M is N mineralized (kg N m^{-2}) in time Δt (s) at optimum water content; M_o (kg N m^{-2}) is the potentially mineralizable N at the beginning of the time interval Δt, K_M is the mineralization rate constant (s^{-1}), and $F(f_{ps})$ is a function of soil moisture. Based on the findings of Stanford and Epstein (1974), and Pilot and Patrick (1972), this function is suggested as (Watts and Hanks, 1978):

$$F(f_{ps}) = 1.111 \, f_{ps}; \quad 0.0 \leqslant f_{ps} < 0.9$$

$$F(f_{ps}) = 10.0 - 10.0 \, f_{ps}; \quad 0.9 \leqslant f_{ps} \leqslant 1.0 \quad \ldots\ldots\ldots [8]$$

where f_{ps} is the fraction of pore space containing water. The value of the mineralization rate constant is computed as a fraction of the absolute soil temperature as follows (Watts and Hanks, 1978):

$$K_M = \exp[17.753 - 6350.5/(T_s + 273)]/(604{,}800) \quad \ldots\ldots [9]$$

where K_M is the mineralization rate constant (s^{-1}) and T_s is the soil temperature (°C). If the soil temperature exceeds 35°C, the rate constant is obtained from the 35°C value (Stanford et al., 1973).

The first order rate equation for nitrification is

$$N_a = [N_{oa}(1 - \exp(-K_N \Delta t))]\, F(f_{ps}) \quad \ldots\ldots [10]$$

where N_a is the amount of ammonium (kg NH_4 m^{-2}) converted to nitrate in time Δt (s) at optimum soil moisture, N_{oa} (kg NH_4 m^{-2}) is the available amount of ammonium subject to conversion at the beginning of time interval, and K_N is the nitrification rate constant (s^{-1}). A water correction factor similar to that introduced for mineralization is used (equation [8]). The nitrification rate constant may be calculated from (Watts and Hanks, 1978):

$$K_N = (0.0105\, T_s + 0.00095\, T_s)K_{N35};\ 0°C < T_s < 10°C$$

$$K_N = (0.032\, T_s - 0.12)K_{N35};\ 10°C < T_s < 35°C$$

$$K_N = (-0.1\, T_s + 4.5)K_{N35};\ 35°C < T_s < 45°C \quad \ldots\ldots [11]$$

where K_N is the nitrification rate constant (s^{-1}), T_s is the soil temperature (°C) and K_{N35} is the rate constant at 35°C (s^{-1}).

Denitrification is computed in the same fashion as nitrification:

$$D = D_0\,(1 - \exp(-K_D \Delta t)) \quad \ldots\ldots [12]$$

where D is denitrification (kg NO_3 m^{-2}) in time Δt, D_o (kg NO_3 m^{-2}) is the available amount of nitrate subject to denitrification at the beginning of the time interval, and K_D (s^{-1}) is the denitrification rate constant. This constant is corrected for water content and temperature as follows (Greene, 1983):

$$K_D = \exp(0.08\,(T_s - 15))K_{D15}\, F(\theta);\ T_s \geqslant 10°C \quad \ldots\ldots [13]$$

$$K_D = 0.67 \exp(0.43(T_s - 10))K_{D15}\, F(\theta);\ T_s < 10°C$$

where K_{D15} is the rate constant at 15°C, T_s is soil temperature and $F(\theta)$ is water content correction function for denitrification, defined as:

$$F(\theta) = \exp(0.304 + 2.94\,(\theta_s - \theta) - 47\,(\theta_s - \theta)^2) \quad \ldots [14]$$

where θ and θ_s are current and saturated water content by volume.

Nitrogen Uptake by Plants

Fig. 2 presents a conceptual diagram for this process, which occurs independently in each node of the soil profile. It is assumed that N flows radially from the bulk soil to the rhizosphere, where the uptake takes place. The rhizophere is composed of nonintersecting, coaxial, cylindrical regions surrounding the roots, whose inner radius is at the root surface and outer radius at some small distance from the root surface where root hairs are active.

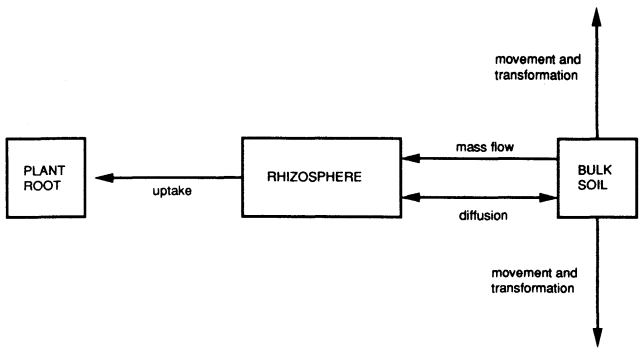


Fig. 2—Model for nitrogen uptake by roots.

Transport of nitrogen from bulk soil to the rhizosphere occurs by convection and diffusion, described as follows:

$$F_N = EN_b + 2\pi\,(N_b - N_r)\,\theta\,\rho_w\,L\,Z\,D_N/(\ln(x/(r_r + r_s))) \quad \ldots\ldots [15]$$

The first term on the right represents convective flow, and the second describes cylindrical diffusion. F_N is nitrogen flux from bulk soil to rhizosphere (kg m^{-2} s^{-1}), E is water uptake (kg H_2O m^{-2} s^{-1}), N_b is bulk soil solution nitrogen concentration (kg N kg^{-1} H_2O), D_N is diffusion coefficient of nitrogen in the soil (m^2 s^{-1}), r_r is root radius (m), r_s is rhizosphere thickness (m), x is distance from root center to point of diffusion (m) assumed to be 1 mm outside of the rhizosphere, N_r is rhizosphere solution nitrogen concentration (kg N kg^{-1} H_2O), θ is volumetric water content (m^3 m^{-3}), p_w is water density (kg m^{-3}), L is root density (m root m^{-3} soil), and Z is soil layer thickness (m). This equation is solved independently for nitrate and ammonium.

The effect of the soil matrix and changes in soil water content on the diffusion coefficient of nitrogen are calculated using (Papendick and Cambell, 1980):

$$D_N = 2.8\, D_{Nw}\, \theta^3 \quad \ldots\ldots [16]$$

where D_{Nw} is the diffusion coefficient for NH_4 or NO_3 in water and θ is volumetric content. The diffusion coefficient in water is given the same value for nitrate and ammonium, as suggested by Atkins (1978).

Nitrogen uptake for each node is simulated using a Michaelis -Menten equation, based on the theory of Nye and Tinker (1977), who consider nitrogen uptake by plants to be dependent on nutrient concentration in the rhizosphere and rate of nutrient uptake per unit length of root.

$$U_N = U_m\, L\, Z\, \Delta t\, F(T_s)\,(1 - R_N/R_{Nm})N_r/(N_r + K_m) \quad \ldots\ldots [17]$$

where U_N is nitrogen uptake (kg N m^{-2}), U_m is maximum

nitrogen uptake per unit of root length (kg N m^{-1} root s^{-1}), L is root density (m root m^{-3} soil), Z is the soil layer thickness (m), Δt is the time interval (s), $F(T_s)$ is a soil temperature function for nitrogen uptake, R_N is the nitrogen reserve pool in the plant (kg N m^{-2}), R_{Nm} is the maximum size of the nitrogen reserve pool in the plant (kg N m^{-2}), N_r is nitrogen concentration in the rhizosphere (kg N kg^{-1} water), and K_m is the saturation constant (kg N kg^{-1} water). Nitrogen and ammonium are combined in equation [17], because the values of maximum uptake and K_m for both nitrate and ammonium are similar (Van Veen and Frissel, 1981).

The temperature function for nitrogen uptake takes on values from 0 to 1, according to the following expression (Campbell, 1977):

$$F(T_s) = [(2(T_s + 4)^2 (T_{ou} + 4)^2 - (T_s + 4)^4]/(T_{ou} + 4)^4 \quad \ldots\ldots [18]$$

where T_s is soil temperature and T_{ou} is optimum temperature for nitrogen uptake (°C), assumed to be 25°C.

Nitrogen Balance in the Soil

The nitrogen balance in the soil is affected by the relative rate of processes described earlier. This mass balance for nitrate and ammonium at each node is obtained as:

$$d(N_N)/dt = \text{FLUX IN} - \text{FLUX OUT} + \text{NITRIFICATION} - \text{DENITRIFICATION} - \text{TRANSPORT TO RHIZOSPHERE} \quad \ldots\ldots [19]$$

$$d(N_a)/dt = \text{FLUX IN} - \text{FLUX OUT} + \text{MINERALIZATION} - \text{NITRIFICATION} - \text{TRANSPORT TO RHIZOSPHERE}$$

Plant Growth in Relation to Carbon and Nitrogen Availability

Plant growth is dependent, among other factors, on C and N acquisition. Both processes are affected by water availability. Equations describing photosynthesis and water uptake processes can be found in Stockle and Campbell (1985).

Growth is divided into that during vegetative and flowering stages and that during grain filling. Attainable growth is defined as the product of potential growth (kg m^{-2} day^{-1}) and a temperature function, whose value ranges from 0 to 1. The potential growth is an input parameter to the model which is probably genotype dependent. Shoot, root and grain growth are represented by the following expressions:

$$G_s = G_{sm}\, F(T_a) \quad \ldots\ldots [20]$$

where G_s is attainable shoot growth per unit time and area (kg m^{-2} day^{-1}), G_{sm} is potential shoot growth (kg m^{-2} day^{-1}) and $F(T_a)$ is an air temperature function for shoot growth, and

$$G_r = G_{rm}\, F(\bar{T}_s)\, F(\bar{\psi}_m) \quad \ldots\ldots [21]$$

where G_r is attainable root growth per unit time and area (kg m^{-2} day^{-1}), G_{rm} is potential root growth (kg m^{-2} day^{-1}), $F(T_s)$ is a root depth-averaged soil temperature function for root growth, and $F(\psi_m)$ is a root depth-averaged water potential function for root growth, and

$$G_g = G_{gm}\, n_g\, G(T_a) \quad \ldots\ldots [22]$$

where G_g is attainable grain growth per unit time and area (kg m^{-2} day^{-1}), G_{gm} is potential growth per grain (kg $grain^{-1}$ day^{-1}), n_g is the number of grain set per unit area (grain m^{-2}) and $G(T_a)$ is air temperature function for grain growth.

The air temperature function for shoot growth was adapted from Rickman et al. (1975):

$$F(T_a) = \exp\,[-(1.023^{Ta}((T_{os} - T_a)/T_a)^2)] \quad \ldots\ldots [23]$$

where T_a is air temperature (°K) and T_{os} is optimum temperature for growth (°K). A value of 25°C was used for optimum temperature (Friend, 1966). The temperature function for root growth is computed in a similar way, using a root depth-averaged soil temperature instead of air temperature. The optimum temperature used was 20°C. Shoot growth appears to have a higher optimum temperature than root growth (Evans et al., 1975).

The water function used for root growth is:

$$\begin{aligned} F(\bar{\psi}_m) &= 1 && ; \bar{\psi}_m > -500\ \text{J kg}^{-1} \\ F(\bar{\psi}_m) &= (\bar{\psi}_m + 1500)/500; \\ & && -500\ \text{J kg}^{-1} > \bar{\psi}_m > -1500\ \text{J kg}^{-1} \quad \ldots\ldots [24] \\ F(\bar{\psi}_m) &= 0 && ; -1500\ \text{J kg}^{-1} > \bar{\psi}_m \end{aligned}$$

where $\bar{\psi}_m$ is the average matric potential (J kg^{-1}) for the depth of soil profile occupied by roots. This equations assumes that rate of root growth is unaffected when $\bar{\psi}_m$ is above -500 J kg^{-1} decreases linearly with $\bar{\psi}_m$ between -500 and -1500 J kg^{-1}, and ceases at -1500 J kg^{-1}.

The temperature function for wheat grain growth is computed as (Maas and Arkin, 1980):

$$\begin{aligned} G(T_a) &= 1 && ; T_a \geqslant 20^\circ\text{C} \\ G(T_a) &= (T_a - 3.0)/17.0; && 3^\circ\text{C} < T_a < 20^\circ\text{C} \quad \ldots\ldots [25] \\ G(T_a) &= 0 && ; T_a \leqslant 3^\circ\text{C} \end{aligned}$$

where T_a is air temperature (°C).

Once the attainable growth for the given environmental conditions is defined, it is expressed in terms of C and N demand:

$$C_s = f_{cs}G_s$$

$$C_r = f_{cr}G_r \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots [26]$$

$$C_g = f_{cg}G_g$$

where C_s, C_r, and C_g are daily carbon demand for shoot, root and grain growth respectively (kg m^{-2} day^{-1}) and f_{cs}, f_{cr}, and f_{cg} are the respective carbon fractions. To compute the N demand, the C/N ratio is required:

$$N_s = C_s/C_{Ns}$$

$$N_r = C_r/C_{Nr} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots [27]$$

$$N_g = C_g/C_{Ng}$$

where N_s, N_r and N_g are daily N demand for shoot, root and grain growth respectively (kg m^{-2} day^{-1}), C_{Ns}, C_{Nr} are C/N ratio for root and shoot respectively, and C_{Ng} is C/N ratio for grain. The C/N ratio for root and shoot growth may be obtained based on observed C/N ratio distributions through the season of low nitrogen content plants. Values of C/N ratios for grain growth must fall within the limits set by the seed protein content of the genotype and variations within the limits can be obtained as a function of current plant nitrogen concentration.

Once the C and N demand has been established, available C and N must be allocated. For this purpose, a mass balance is performed:

$$d(R_C)/dt = P_n - C_sR_CR_N/[(K_{Cs} + R_C)(K_{Ns} + R_N)]$$

$$- C_rR_CR_N/[(K_{Cr} + R_C)(K_{Nr} + R_N)]$$

$$- C_gR_CR_N/[(K_{Cg} + R_C)(K_{Ng} + R_N)] \quad \ldots\ldots [28]$$

where the term on the left is the change in time of the C reserve pool, the first term on the right is net photosynthetic rate, the second is the rate of C allocation for shoot growth, the third term is the rate of C allocation for root growth, and the last is the rate of C allocation for grain growth. The change in N reserve is given by

$$d(R_N)/dt = U_{Nt} - N_sR_CR_N/[(K_{Cs} + R_C)(K_{Ns} + R_N)]$$

$$- N_rR_CR_N/[(K_{Cr} + R_C)(K_{Nr} + R_N)]$$

$$- N_gR_CR_N/[(K_{Cg} + R_C)(K_{Nr} + R_N)] \quad \ldots\ldots [29]$$

where the first term on the right is N uptake rate, the second is the rate of N allocation for shoot growth, the third term is the rate of N allocation for root growth, and the last one is the rate of N allocation for grain growth. A more detailed partition of dry matter is possible, where more terms must be added to equations [28] and [29].

A finite difference approach using a fully backward scheme and daily time step was used to solve the differential equations for C and N allocation. Equations [28] and [29] must be solved simultanously, and because the unknown variables, R_N and R_C, appear in the storage terms (left hand side of the equations) and in the allocation rate terms, iteration is needed. The Newton-Raphson method was used for this purpose. In the equations, P_n is net photosynthetic rate (kg m^{-2} day^{-1}), U_{Nt} is N uptake rate (kg m^{-2} day^{-1}), R_C is the C reserve pool (kg m^{-2}), R_N is the N reserve pool (kg m^{-2}), K_{Cs}, K_{Cr}, and K_{Cg} are the C saturation constants for shoot, root and grain respectively, and K_{Ns}, K_{Nr}, and K_{Ng} are the N saturation constants (kg m^{-2}). K values are input parameters indicating priority in the allocation of C or N between the three sinks.

Based on the cumulative dry matter allocation to shoots and roots, leaf area index and root growth are computed respectively (Stockle and Campbell, 1985).

C and N taken up in excess of demand go to reserve pools, whose sizes need to be defined. The C reserve pool can be defined as a percent of the total biomass carbon. The size of the N reserve pool can be approximated based on the difference between maximum and minimum observed plant N concentration. Photosynthesis is inhibited when the C reserve pool is at maximum. This is an indirect control over photosynthetic rate, which would be more properly included through changes in the internal CO_2 concentration at the stomates. Nitrogen uptake is also inhibited when the reserve pool is at maximum (see N uptake section).

Grain Number

A potential grain number per unit area is obtained as the product of the number of spikes per unit area times the number of grains per spike for an unstressed crop. This value is genotype dependent. The potential grain number is corrected by a water and a nitrogen stress index. The water stress index (WSI) has been defined as (Hiler and Clark, 1971)

$$S_W = \Sigma(1 - T/T_p) \quad \ldots\ldots\ldots\ldots\ldots\ldots [30]$$

where T is actual transpiration (kg m^{-2} s^{-1}) and T_p is potential transpiration (kg m^{-2} s^{-1}). The nitrogen stress index (S_N) is defined in a similar fashion:

$$S_N = \Sigma(1 - N_p/(N_{pm}f_s)) \quad \ldots\ldots\ldots\ldots\ldots\ldots [31]$$

where f_s corresponds to the fraction of maximum plant nitrogen concentration below which stress would occur. These indeces are independently accumulated for each growth stage.

The mechanism of the effect of water and nitrogen stress on the number of grains set is not well understood. For wheat, a few field experiments provide preliminary quantification of this relation (Choudhury and Kumar, 1980; Robins and Domingo, 1962; Shimshi and Kafkafi, 1978; Parameswaran et al., 1984; Singh and Malik, 1983). Based on these references, correction factors accounting for the effect on grain number of water and nitrogen stress at different growth stages were obtained. This stress effect can be either on number of spikes per square meter or number of grains per spike or both. The approach described is highly empirical and a better simulation of grain number should be developed for inclusion in the model. Population density is a factor determining grain number per unit area that was not considered. Detailed simulations of tillering an competition should also be developed.

Grain Filling and Translocation

Grain growth during the filling period is the result of production of assimilates and storage of carbon and nitrogen compounds in the developing grains. All the C and N resources are allocated for grain growth during this period. However, the increase in grain weight often exceeds dry matter production after anthesis, sometimes by 50% or more. When photosynthesis after anthesis is slow and senescence is fast, the resulting shortfall in assimilate for grain growth is compensated by translocation (Gallagher and Biscoe, 1978). The level of available nitrogen in the soil is usually low during grain filling. Thus, the nitrogen requirement of the developing grains is largely satisfied by translocation (Simpson et al., 1983). Spiertz (1977) indicates that the losses of carbohydrates and nitrogen from stems and leaves of wheat amounted to about 30% and 60% of the ultimate quantities of carbon and nitrogen compounds in the grains respectively. The mechanism of translocation is not well understood. Furthermore, differences in ability of wheat genotypes to translocate have been reported (Blum et al., 1983). Translocation has been included in the model defining a daily percent of non-grain C and N that is made available for grain filling.

CALIBRATION AND TESTING

The conceptual model was incorporated into a computer simulation model for predicting spring wheat yield in response to water and nitrogen. This model was calibrated with data collected in 1983 from a field experiment on a Hanning silt loam soil located near Davenport, WA and tested with data collected in 1983 from a line source experiment in Logan, UT. The model is written in Pascal, and a version for the IBM PC or any compatible microcomputer is available upon request.

Calibration

Model calibration consisted of adjustments of biological parameters within the range of fluctuation reported in the literature to fit the limited experimental data collected. Physical parameters were independently estimated and were not adjusted.

On April 29, 1983, Dirkwin spring wheat was seeded in plots 1.8 m wide x 12.0 m long. Six rows at a spacing of 0.3 m were planted in each plot. Plots were harvested on August 29. Two levels of nitrogen supply, 0 and 101 kg N/ha, with three replications were used. Air temperature, soil temperature at 3 and 50 cm depth, solar radiation, wind speed and rainfall were measured. Temperatures were measured using copper-constantan thermocouples, solar radiation with a silicon cell pyranometer, wind speed with a cup anemometer, and rainfall with a tipping bucket rain gage. Sensors were scanned every minute and averaged over six hour periods using a data logger (Model CR5, Campbell Scientific, Inc., Logan, UT).

Soil water content to a depth of 150 cm was monitored through the season in 15 cm depth increments with a neutron probe. Water content in the top 7.5 cm of the soil profile was determined gravimetrically.

Dry matter accumulation and leaf area index were measured on each plot through the season. Plant samples were collected by removing all above ground plant parts within a meter row section. Leaf area index was determined using a leaf area meter (model 3100, LiCor, Inc, Lincoln, NE). Above ground plant material was dried at 70°C for 72 hours and weighed. Plant nitrogen content was obtained from data reported by Parsons (1984).

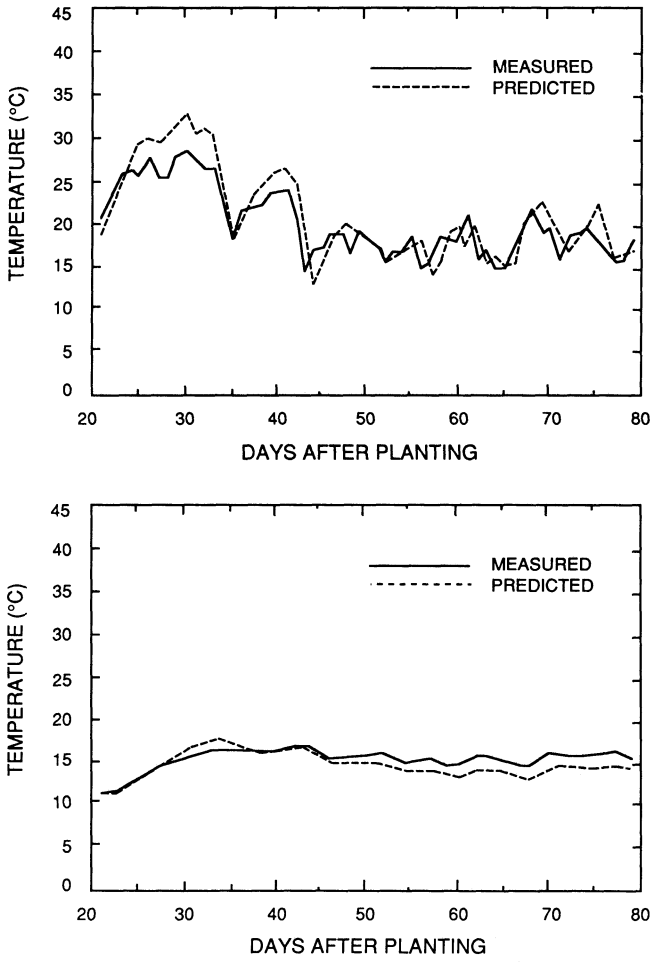


Fig. 3—Measured and predicted soil temperature (Davenport) (a) at 3 cm depth (above), (b) at 50 cm depth (below).

Fig. 3 shows a simulation of soil temperature from 20 to 80 days after planting at 3 and 50 cm depth. Simulated values compare well with measured ones. There is more error in the soil surface simulations than deeper in the profile. The model tends to overpredict soil temperature at the beginning of the season, when LAI is low, and to underpredict when the canopy become denser. The temperature predictions are probably accurate enough for modeling rates of biological processes in the soil.

A comparison of simulated and predicted water content at three times during the season for the 101 kg/ha treatment is shown in Fig. 4. Model predictions agree well with the measured values. Similar results were obtained with the 0 kg/ha treatment.

Simulated and measured values of leaf area index are presented in Fig. 5. LAI of the fertilized treatment was well simulated, but LAI for the unfertilized treatment was overpredicted, explained partially by overprediction of dry matter during that period (Fig. 6), and may be indicating inaccuracy in the LAI-dry matter relationship used to predict LAI.

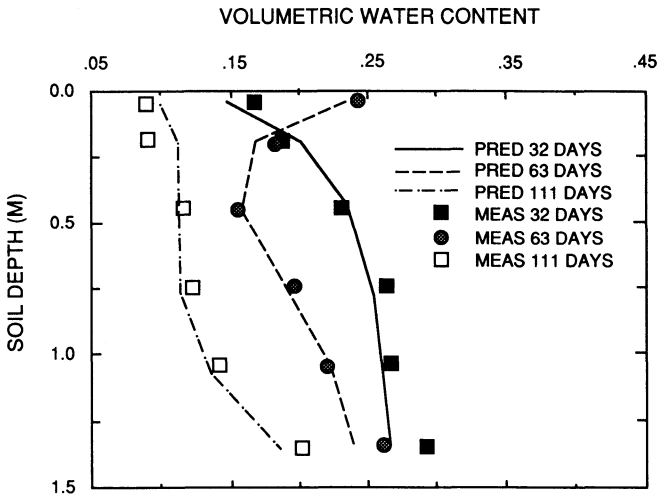


Fig. 4—Measured and predicted volumetric soil moisture content at three times for the 101 kg N/ha treatment (Davenport).

Simulated top dry matter accumulation (stem, leaves and grain) and plant nitrogen content through the season are shown in Figs. [6] and [7], respectively. Simulation of dry matter follows the seasonal trend and compares well with measured data. The good agreement between predicted and measured plant nitrogen content suggest that the model simulates reasonably nitrogen uptake by the crop. Grain yields (3.5 and 1.3 ton/ha for 101 and 0 kg N/ha, respectively) were closely predicted.

Testing the Model

The calibrated model was tested using data collected by Reynolds (1983) in Logan, UT, consisting of spring wheat dry matter and yield response to six levels of applied water ranging from 28 to 50 cm, and five nitrogen levels, ranging from 0 to 228 kg/ha.

A comparison of predicted and simulated total dry matter and grain yield is shown in Fig. 8. The trend of grain yield was well simulated, but all the values were underpredicted. Total dry matter was also underpredicted, particularly at high yield. Several factors, both biological and physical, could be responsible for underprediction of yield.

A sensitivity analysis showed that important model

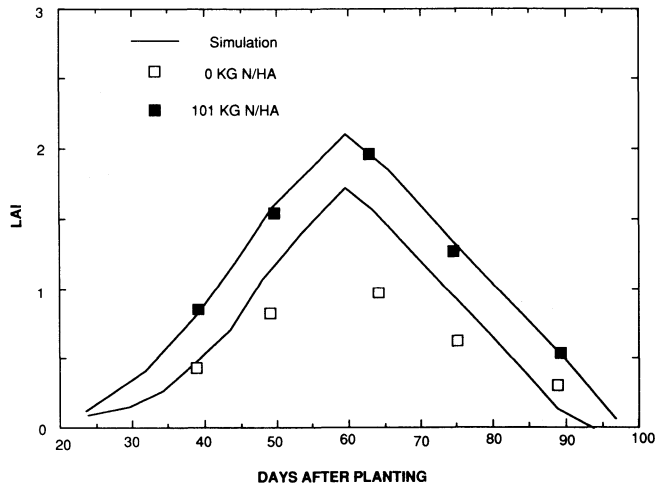


Fig. 5—Measured and predicted leaf area index (Davenport).

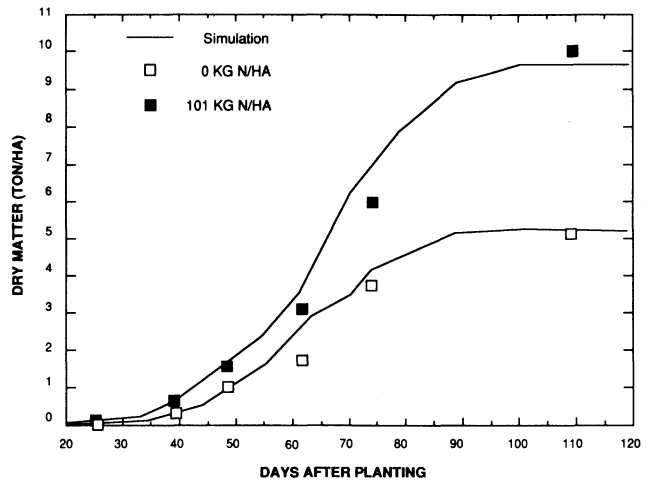


Fig. 6—Measured and predicted total dry matter production (Davenport).

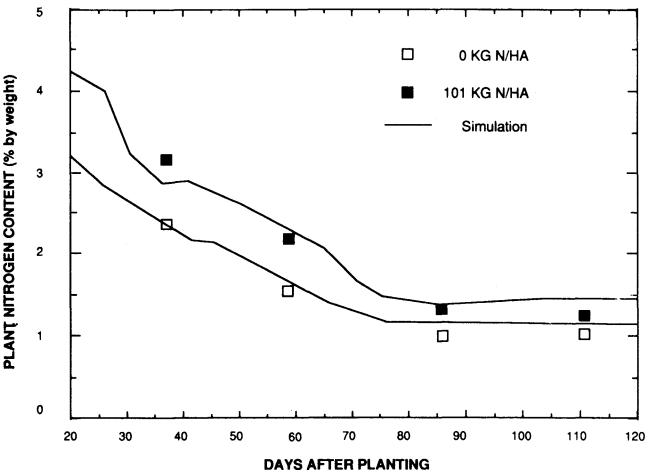
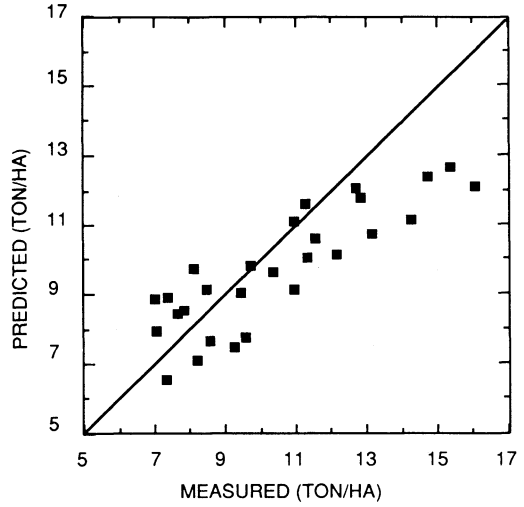


Fig. 7—Measured and predicted plant nitrogen content (Davenport).

outputs (e.g., total dry matter, grain yield, evapotranspiration, and N uptake) were, in general, more sensitive to variation in biological parameters than to variation in physical parameters. This is likely the case in the field as well. Biological parameters are difficult to quantify and subject to greater variability than physical parameters.

Because most of the N uptake occurs before grain filling, N can be a limiting factor for grain production if not enough translocated N is available. Translocated C availability can also be a limiting factor. The translocation rate permitted in the original calibration may not aply to the wheat variety grown at Logan. Differences in ability to translocate between different varieties have been documented (Blum et al., 1983; Austin et al., 1977; Rawson and Evans, 1971). N translocation rate, and also C translocation rate, to a minimum extent, were increased and new simulations were performed (Fig. 9). Agreement between measurements and simulation was improved and the scatter was reduced. Note that both grain yield and total dry matter were increased, which is the result of increased C fixation during grain filling. Larger N availability permitted larger C consumption for production so that the C reserve pool was more depleted,



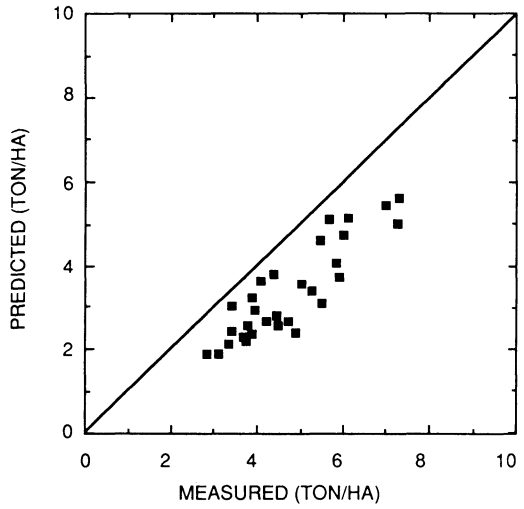
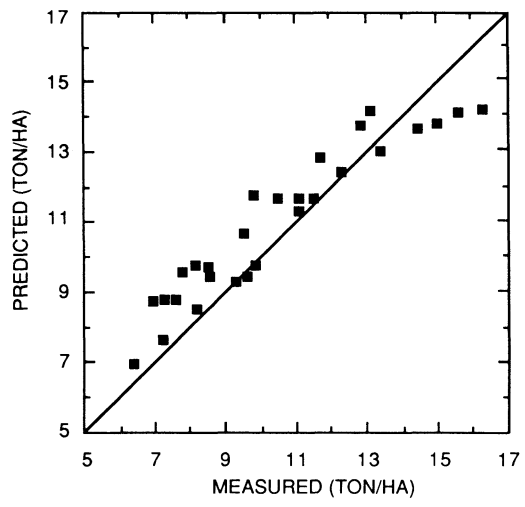


Fig. 8—Measured and predicted (a) dry matter production in ton/ha (above) and (b) grain yield in ton/ha (below) at Logan, using original translocation parameter.



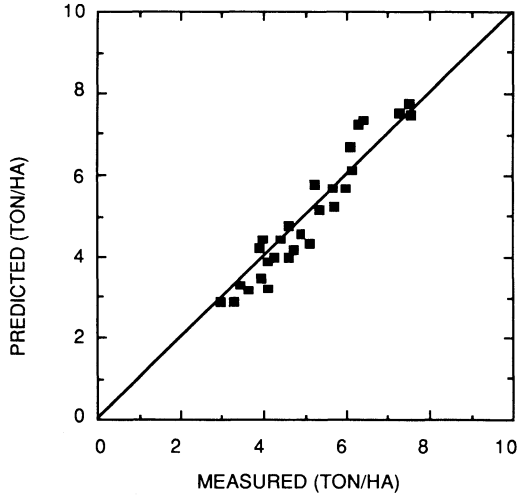


Fig. 9—Measured and predicted (a) dry matter production in ton/ha (above) and (b) grain yield in ton/ha (below) at Logan, using adjusted translocation parameter.

diminishing feedback inhibition of photosynthesis. Other factors that could cause variation (e.g., photosynthetic response to light, grain number sensitivity to stress, etc.) were not analyzed. However, the conclusion seems to be that values of biological parameters that are cultivar dependent must be known for groups of cultivars to improve simulation capabilities. The conceptual model presented includes a reasonably detailed treatment of physical processes with somewhat less emphasis on the treatment of biological processes. Future efforts should strengthen the crop growth description.

SUMMARY AND CONCLUSIONS

A model for simulating the effect of water and nitrogen on growth and yield of spring wheat was discussed. The model was calibrated against data collected at Davenport, WA showing good agreement between measurements and simulations of soil temperature, water content, leaf area index, dry matter accumulation and plant nitrogen content for two nitrogen treatments (0 and 101 kg N/ha). Testing of the calibrated model with data from Logan, UT suggested that C and N translocation functions might need to be adjusted for different cultivars. A larger data set is required for a more rigorous calibration and validation of the model.

The model attempts to offer a mechanistic integration of water, heat and solute transport, nitrogen transformations, and nitrogen and water uptake for the simulation of crop growth and yield. Crop growth, although treated with a reasonable level of detail, remains empirical in many aspects. A mechanistic description of tillering, grain numbers, expansive growth, and translocation needs to be incorporated to improve model capabilities.

References

1. Atkins, P. W. 1978. *Physical Chemistry.* W. H. Freeman Co. San Francisco.

2. Austin, R. B. et al. 1977. The fate of dry matter carbohydrates and C^{14} loss from leaves and stems of wheat during grain filling. *Ann. Bot. London,* 41:1309-1321.

3. Baker, D. N. 1981. Winter wheat: A model for the simulation of growth and yield in winter wheat. AGRISTAR, Yield Model Development. YM-U2-04281.

4. Blum, A. et al. 1983. Chemical dessication of wheat plants as a simulator of post-anthesis stress. I. Effects on translocation and kernel

growth. *Field Crop Research* 6:51-58.

5. Bristow, K. L. 1982. Simulation of heat and moisture transfer through a surface residue-soil system. Ph.D. thesis. Washington State University, Pullman.

6. Campbell, G. S. 1977. *An introduction to environmental biophysics.* New York: Springer Verlag.

7. Campbell, G. S. 1985. Soil phisics with BASIC: Transport models for soil-plant systems. Elsevier, Amsterdam.

8. Choudhury, P. N. and V. Kumar. 1980. The sensitivity of growth and yield of dwarf wheat to water stress at three growth stages. *Irrig. Sci.* 1:223-231.

9. Evans, L. T., I. F. Wardlaw and R. A. Fischer. 1975. Wheat. In *Crop physiology,* L. T. Evans, ed. London: Cambridge University Press.

10. Friend, D. J. C. 1966. The effects of light and temperature on the growth of cereals. In *The growth of cereals and grasses.* F. L. Milthorpe and J. D. Ivans, eds., Butterworth, London.

11. Gallagher, J. N. and P. V. Biscoe. 1978. A physiological analysis of cereal yield. II. Partitioning of dry matter. *Agricultural Progress* 53:51-70.

12. Greene, J. 1983. Simulation of denitrification in a forest soil. M.S. thesis. Washington State University, Pullman.

13. Hiler, E. A. and R. N. Clark. 1971. Stress day index to characterize effects of water stress on crop yields. *Transactions of the ASAE* 14(4):757-761.

14. Jones, C. A. and J. R. Kiniry. 1986. CERES-Maize, a simulation model of maize growth and development. College Station: Texas A & M University Press.

15. Keulen, van, H. and N. G. Seligman. 1987. Simulation of growth, water use and nitrogen nutrition of spring wheat. Simulation Monographs, PUDOC, Wageningen (in press).

16. Maas, S. J. and G. F. Arkin. 1980. *TAMW: A wheat growth and development simulation model.* Texas Agric. Exp. Sta., Blackland Research Center, Temple, TX.

17. Mehran, M. and K. K. Tanji. 1974. Computer modeling of nitrogen transformation in soils. *J. Environ. Quality* 3(4):391-396.

18. Meyer, G. E. et al. 1979. A dynamic simulator of soybean growth, development and seed yield: I. Theory, structure and validation. Research Bulletin of the Ohio Agricultural Research and Development Center, No. 1113.

19. Nye, P. H. and P. B. Tinker. 1977. *Solute movement in the soil-root system.* Oxford: Blackwell.

20. Papendick, R. I. and G. S. Campbell. 1980. Theory and measurement of water potential. In *Water potential relations in soil microbiology,* p. 1-21. Madison, WI: Soil Sci. Soc. Am.

21. Parameswaran, K. V. M., R. D. Graham and D. Aspinall. 1984. Studies on the nitrogen and water relations of wheat. II. Effects a varying nitrogen and water supply on growth and grain yield. *Irrig. Sci.* 5:105-121.

22. Pilot, L. and W. H. Patrick, Jr. 1972. Nitrate reduction in soils: Effect of soil moisture tension. *Soil Sci.* 114:213-216.

23. Rawson, H. M. and L. T. Evans. 1971. The contribution of stem reserves to grain develpment in a range of wheat cultivars of different height. *Aust. J. Agric. Res.* 22:851-863.

24. Reynolds, E. D. 1983. Nitrogen, water, variety, interactions of wheat. M.S. thesis. Utah State University, Logan.

25. Rickman, R. W., R. E. Ramig and R. R. Allmaras. 1975. Modeling dry matter accumulation in dryland winter wheat. *Agron. J. 67:283-289.*

26. Robins, J. S. and C. E. Domingo. 1962. Moisture and nitrogent effects on irrigated spring wheat. *Agr. J.* 54:135-138.

27. Shimshi, D. and U. Kafkafi. 1978. The effect of supplemental irrigation and nitrogen fertilization on wheat. *Irrig. Sci.* 1:27-38.

28. Simpson, R. J., H. Lambers and M. Dalling. 1983. Nitrogen redistribution during grain growth in wheat. IV. Development of a quantitative model of the translocation of nitrogen to the grain. *Plant Physiol.* 71:7-14.

29. Singh, T. and D. S. Malik. 1983. Effect of water stress at three growth stages on the yield and water use efficiency of dwarf wheat. *Irrig. Sci.* 4:239-245.

30. Spiertz, J. H. J. 1977. The influence of temperature and light intensity on grain growth in relation to the carbohydrate and nitrogen economy of the wheat plant. *Neth. J. Agric. Sci.* 25:183-197.

31. Stanford, G. and S. J. Smith. 1972. *Nitrogen mineralization potentials of soils.* Soil Sci. Soc. Amer. Proc. 36:465-472.

32. Stanford, G. and E. Epstein. 1974. *Nitrogen mineralization-water relations in soils.* Soil Sci. Soc. Am. Proc. 38:103-107.

33. Stockle, C. O. 1985. Simulation of the effect of water and nitrogen stress on growth and yield of spring wheat. Ph.D. thesis, Washington State University, Pullman.

34. Stockle, C. O. and G. S. Campbell. 1985. A simulation model for predicting effect of water stress on yield: An example using corn. *Advances in Irrigation,* 3:283-311.

35. Thom, A. S. 1975. Momentum, mass and heat exchange of plant communities. In *Vegetation and the atmosphere,* Vol. I, J. L. Monteith, ed. London: Academic Press.

36. Tillotson, W. R. and R. J. Wagenet. 1982. Simulation of fertilizer nitrogen under cropped situations. *Soil Sci.* 133(3):133-143.

37. Van Veen, J. A. and M. F. Frissel. 1981. *Simulation of Nitrogen Behavior of Soil-Plant Systems,* p. 126-144. Centre for Agriculture Publishing and Documentation, Wageningen, Netherlands.

38. Watts, D. G. and R. J. Hanks. 1978. A soil-water-nitrogen model for irrigated corn on sandy soils. *Soil Sci. Soc. Am. J.* 42:492-499.